Deep Learning for Image Classification

Comparison of AlexNet and DHVT

on the Same Image Classification Dataset

Report

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Date:	March 31, 2026

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Abstract

This report presents a comparative study of AlexNet and DHVT under the same dataset setting. The goal is to compare a classical convolutional neural network with a transformer-style architecture using the same data configuration, evaluation pipeline, and custom test images. The report introduces the dataset, explains the structure of both models, summarizes the training procedure, discusses the main implementation challenges, and analyzes the visual evidence used to compare the two approaches.

1 Introduction

Image classification is one of the most fundamental tasks in computer vision. A classification model receives an input image and predicts the most likely class among a predefined set of categories. In this project, the aim is not only to train a model, but also to compare two different families of architectures: convolutional neural networks and transformer-based models.

The first model is AlexNet, a historically important convolutional neural network that played a major role in the development of deep learning for vision. The second model is DHVT, a transformer-style architecture designed for small datasets. Comparing these two models on the same dataset makes it possible to study the difference between a classical CNN and a more recent token-based architecture under a fairer and more controlled setting.

2 Project Objectives

The main objectives of this project are summarized in Table 1.

No.	Objective
1	Train an AlexNet-based model on a selected image classification dataset.
2	Train a DHVT model on the same dataset.
3	Save and reload the trained weights for both models.
4	Run inference on personal images stored in a custom test folder.
5	Compare the two models under the same dataset configuration.

Table 1: Main project objectives.

3 Why Compare AlexNet and DHVT

AlexNet is a classical CNN architecture that relies on convolution, pooling, and fully connected layers. It is easier to interpret from a traditional computer vision perspective because it extracts local features progressively through stacked convolutional layers.

DHVT, on the other hand, processes an image through token embeddings and attention-based interactions while still keeping inductive bias for small datasets. This makes it conceptually different from CNNs and particularly interesting for comparison.

This comparison is useful because it highlights the differences summarized in Table 2.

Aspect	Comparison meaning
Classical baseline	AlexNet represents a convolutional architecture with strong inductive bias for images.
Modern architecture	DHVT represents a transformer-style model with attention-based token interactions.
Training behavior	The two models may respond differently when trained on relatively small or medium datasets.

Table 2: Why the comparison between AlexNet and DHVT is meaningful.

At the same time, a fair comparison should not describe only the strengths of each family. AlexNet is simple and interpretable, but it may struggle to capture long-range dependencies and can become parameter-heavy because of its large classifier layers. Transformer-style models such as DHVT can model more global interactions, but they are usually more complex to implement, often require more careful optimization, and can be more sensitive to data scale, regularization, and hyperparameter choice than classical CNN baselines. Therefore, the purpose of the comparison is not to claim that one family is universally superior, but to examine the practical trade-offs under the same experimental conditions.

4 Dataset

The implementation is designed to support multiple datasets through the same notebook configuration. The supported options are summarized in Table 3.

Dataset option	Description
`CIFAR-10`	Standard 10-class dataset used for baseline comparison.
`CIFAR-100`	Fine-grained 100-class dataset with the same image format.
`custom_folder`	Folder-based dataset using the structure `train/class_name/...` and `val/class_name/...`.

Table 3: Dataset options supported by the notebook.

This flexibility is important because it allows the same comparison notebook to be reused without changing the global experimental structure. For a fair comparison, both AlexNet and DHVT are trained and evaluated on the same dataset choice.

5 Model Description

5.1 AlexNet

AlexNet is a deep convolutional neural network composed of several convolutional layers followed by pooling operations and fully connected layers. In this project, the AlexNet-style model includes:

- convolutional layers for feature extraction,

- ReLU activations for non-linearity,
- max-pooling layers for spatial reduction,
- adaptive average pooling before the classifier,
- fully connected layers for final classification.

AlexNet is well suited for educational purposes because its structure is clear and interpretable. Even though it is an older architecture, it remains useful as a strong baseline for image classification tasks.

5.1.1 Mathematical view of convolution

For an input feature map X and a convolution kernel K, the output at position (i, j) can be written as:

$$Y_{i,j} = \sum_u \sum_v K_{u,v} \cdot X_{i+u,j+v}.$$

This operation allows the network to detect local patterns such as edges, textures, and shapes. After convolution, the non-linear activation is applied:

$$\mathrm{ReLU}(x) = \max(0, x).$$

Pooling then reduces the spatial resolution while preserving the strongest responses.

5.2 DHVT: Dynamic Hybrid Vision Transformer

The main transformer-style model retained for the report is a DHVT-inspired architecture trained from scratch. Its purpose is to better adapt transformer ideas to smaller datasets. The implementation used in the notebook is composed of the following elements:

- a **Shared Overlapped Patch Embedding** (SOPE) module;
- a class token and positional embeddings;
- repeated DHVT blocks;
- head-token based attention;
- a DAFF module used inside each block;
- a final linear classifier.

5.2.1 SOPE module

Instead of directly flattening non-overlapping patches, SOPE uses convolutional layers with stride to create embedded patch tokens. If the input image is denoted by

$$X \in \mathbb{R}^{B \times C \times H \times W},$$

then the SOPE block produces a tensor

$$E \in \mathbb{R}^{B \times N \times d},$$

where N is the number of embedded tokens and d is the embedding dimension.

This approach introduces useful locality before the transformer blocks and is therefore better suited to small datasets than a purely patch-based tokenization.

5.2.2 Class token and positional encoding

After the patch tokens are created, a learnable class token is concatenated:

$$Z_0 = [x_{\text{cls}}; E].$$

Then positional embeddings are added:

$$Z = Z_0 + P,$$

where P is a learnable positional embedding matrix. This gives the model information about the spatial arrangement of the tokens.

5.2.3 DHVT block

Each DHVT block contains a head-token style self-attention layer followed by a DAFF module. The residual structure can be summarized as:

$$Z' = Z + \text{MHSA}(\text{LN}(Z)),$$

$$Z'' = Z' + \text{DAFF}(\text{LN}(Z')).$$

Here, LN is layer normalization and MHSA denotes multi-head self-attention.

5.2.4 Attention mechanism

For query, key, and value matrices, the model uses:

$$Q = ZW_Q, \quad K = ZW_K, \quad V = ZW_V.$$

The attention output is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

This formulation allows the model to capture global relationships between image tokens.

5.2.5 DAFF module

The DAFF module acts as an enhanced feed-forward stage. In the implementation, it includes:

- a first linear projection,
- a depthwise convolution,
- a GELU activation,
- a second linear projection,
- a channel-attention style reweighting for the class token.

If the patch tokens are denoted by T, the DAFF stage can be simplified as:

$$T_1 = W_1 T,$$

$$T_2 = \mathrm{DWConv}(T_1),$$

$$T_3 = W_2 \,\mathrm{GELU}(T_2).$$

Then a channel weight vector w is learned and applied to the class token:

$$w = \sigma(f(\mathrm{mean}(T_3))),$$

$$x'_{\mathrm{cls}} = x_{\mathrm{cls}} \odot w,$$

where σ is the sigmoid function and $\odot$ is element-wise multiplication.

5.2.6 Final classification

After the final DHVT block, the normalized class token is extracted and passed to a linear layer:

$$\hat{y} = W_{\mathrm{head}} x_{\mathrm{cls}}^{(L)} + b.$$

The model is trained with cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{p}_c),$$

where $\hat{p}_c$ is the predicted probability for class c after the softmax function.

6 Experimental Method

The same global workflow is used for both models, as shown in Table 4.

Step	Procedure
1	Choose the runtime environment (local or Colab).
2	Choose the dataset.
3	Preprocess the images.
4	Initialize the model.
5	Train the model.
6	Evaluate it on the validation set.
7	Save the best weights.
8	Reload the saved weights.
9	Run inference on custom images.

Table 4: Shared workflow for AlexNet and DHVT.

To make the comparison fair, the dataset, number of epochs, and evaluation procedure are shared across both models.

7 Preprocessing

Both models use image resizing to 224×224, because this resolution is compatible with the selected architectures. Data augmentation is applied during training to improve generalization. The main preprocessing operations are listed in Table 5.

Operation	Purpose
Horizontal flipping	Increase left-right appearance variation.
Small random rotations	Improve robustness to minor orientation changes.
Tensor conversion	Convert input images into tensors for PyTorch.
Normalization	Stabilize training and keep the input range consistent.

Table 5: Main preprocessing operations used in the notebook.

Different normalization values are used for AlexNet and ViT. AlexNet follows the simpler normalization adopted in the previous notebook, whereas ViT uses ImageNet normalization because the pretrained backbone was originally trained on ImageNet.

8 Training Strategy

For AlexNet, the optimizer is AdamW and the model is trained from scratch on the selected dataset. For DHVT, the model is also trained from scratch, which is consistent with the goal of evaluating a transformer-style architecture adapted to small datasets rather than only relying on transfer learning.

The learning rate scheduler decreases the learning rate during training in order to stabilize optimization. Validation accuracy is monitored after every epoch, and the best model weights are saved separately from the last model weights.

9 Main Difficulties Encountered

Several practical and conceptual difficulties appeared during the project.

9.1 Difficulty 1: Comparing Two Models Fairly

At first, using two different notebooks or two different datasets would have made the comparison less meaningful. If the data are not the same, the final conclusions about AlexNet and DHVT are weaker.

Solution. Both models were reorganized into the same notebook and forced to share the same dataset configuration, the same validation set, and the same custom image folder.

9.2 Difficulty 2: Transformer Training on Small Datasets

One important issue is that transformer-style models often require careful design to remain effective on small datasets. A naive transformer implementation may perform poorly or train very slowly.

Solution. The final implementation uses a DHVT-inspired model with overlapped patch embedding, token-based attention, and a DAFF module. This is closer to the goal of making transformer-style architectures more compatible with small datasets.

9.3 Difficulty 3: Keeping the Work in Homework Style

Another difficulty was to avoid writing the project in a purely engineering style. The notebook had to remain understandable as a class assignment, with a clear step-by-step structure and readable explanations.

Solution. The final implementation was kept in notebook format, with explicit sections, simple code flow, and full textual explanations in English.

9.4 Difficulty 4: Reusability for Other Datasets

If the notebook had been written only for CIFAR-10, it would have been less useful for later experiments.

Solution. A dataset selection section was added, allowing the user to switch between `CIFAR-10`, `CIFAR-100`, and a folder-based custom dataset.

9.5 Difficulty 5: Long Training Time and Missing Intermediate Output

When training DHVT from scratch, especially on Colab, it is possible to wait a long time before seeing any visible progress. This creates confusion because the model may appear frozen even when it is still running.

Solution. The DHVT training code was written with regular progress printing inside the batch loop, so the user can observe the evolution of the training process without waiting for a full epoch before seeing output.

10 Results

The notebook produced a full set of quantitative and qualitative comparisons between AlexNet and DHVT under the same CIFAR-10 setting. The most important observation is that the DHVT implementation is not only more accurate than AlexNet, but also more parameter-efficient and more consistent across classes. Table 6 summarizes the main metrics extracted from the final notebook outputs.

Metric	AlexNet	DHVT
Best validation accuracy	0.6848	0.8062
Number of parameters	58,322,314	6,504,202
Macro precision	0.6811	0.8057
Macro recall	0.6848	0.8062
Macro F1-score	0.6814	0.8056
Expected calibration error (ECE)	0.0211	0.0198

Table 6: Main numerical outputs extracted from the notebook.

Table 6 already supports three important conclusions. First, DHVT achieved a clearly higher best validation accuracy than AlexNet, improving from 68.48% to 80.62%. Second, this

improvement was achieved with far fewer parameters, which makes the result more interesting than a simple capacity comparison. Third, the macro precision, recall, and F1-score all show the same ranking, so the advantage of DHVT is not limited to a single metric.

10.1 Training and Validation Curves

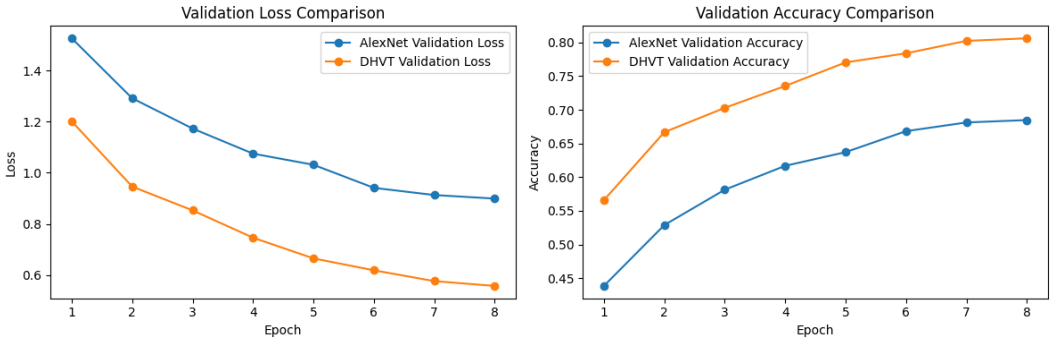

Figure 1: 01. Validation loss and validation accuracy curves of AlexNet and DHVT.

Figure 1 compares the validation loss and validation accuracy of the two models across training epochs. The curves indicate that DHVT learns more effectively than AlexNet on the current dataset. Its validation accuracy rises to a clearly higher level, and its validation loss decreases more strongly. AlexNet remains a meaningful baseline, but the gap in the final validation accuracy suggests that the DHVT design is better suited to the current task.

This figure is especially important because it provides a dynamic view of learning rather than only a final number. It shows whether the model converges quickly, whether the training remains stable, and whether there is any sign of plateauing. In this experiment, DHVT not only reaches a better end result, but also shows a stronger overall trajectory during training.

10.2 Confusion Matrices

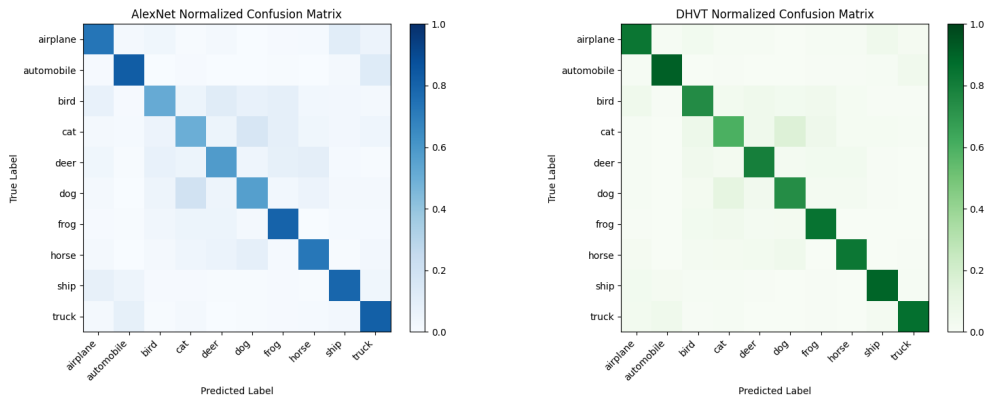

Figure 2: 02. Confusion matrices of AlexNet and DHVT on the evaluation set.

Figure 2 is one of the most informative class-level analyses in the report. The diagonal entries summarize the per-class success rates, while the off-diagonal entries reveal which categories are visually close to each other from the perspective of each model. In the current experiment, both models still confuse semantically similar classes such as cat and dog, but the DHVT

matrix is noticeably more concentrated on the diagonal, which is consistent with its better overall accuracy.

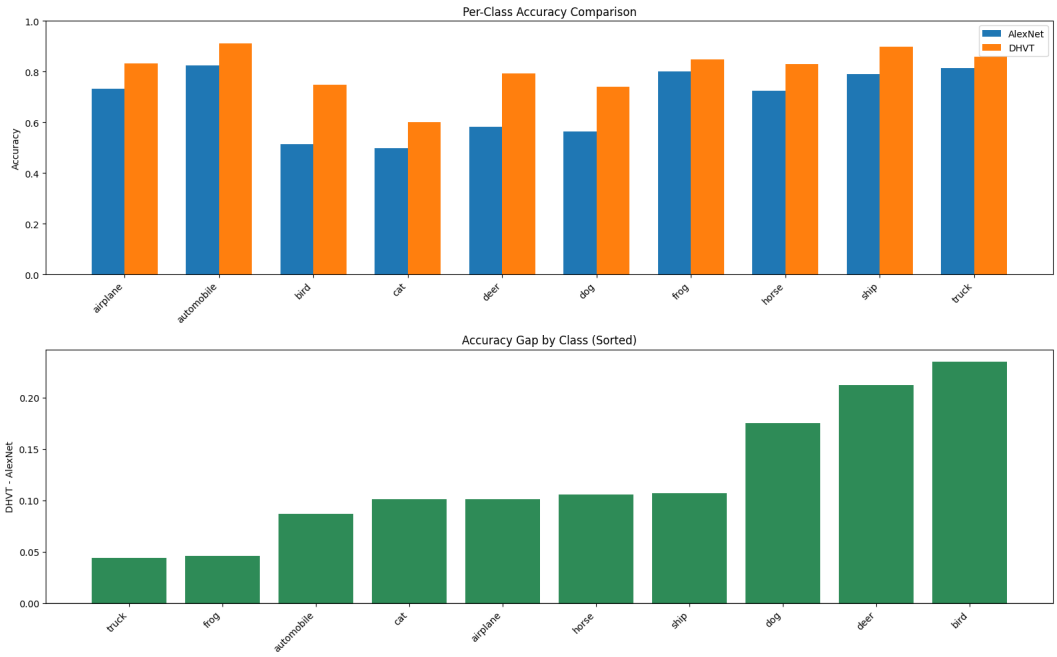

Figure 3: 07. Per-class accuracy comparison between AlexNet and DHVT.

Figure 3 shows that the gain of DHVT is not concentrated in only one or two categories. The transformer-style model improves performance on almost every class, with especially visible gains on visually challenging categories such as bird, deer, and dog. This figure is important because it demonstrates that the higher overall accuracy is the result of broad class-level improvement rather than a narrow specialization.

Another useful detail is that the classes with the largest gains are also classes in which local texture alone is often insufficient. In those categories, the ability of DHVT to combine local and more global information appears to be a practical advantage over the purely convolutional baseline.

10.3 Misclassified Examples

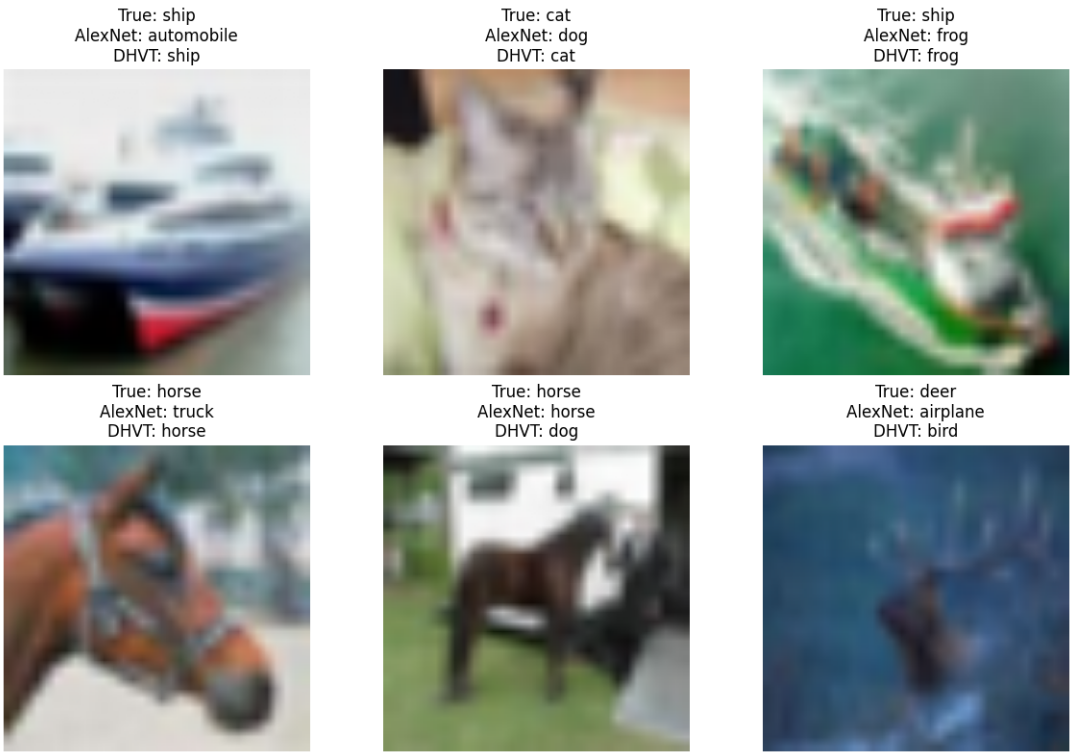

Figure 4: 03. Examples in which at least one of the two models makes an incorrect prediction.

Figure 4 provides a qualitative complement to the numerical metrics. It reveals the kinds of visual situations that remain difficult, such as cluttered backgrounds, unusual object poses, low-resolution details, or strong similarity between classes. It is also useful for identifying cases in which one architecture succeeds while the other fails, which suggests that the two models rely on different visual cues.

10.4 Examples Misclassified by Both Models

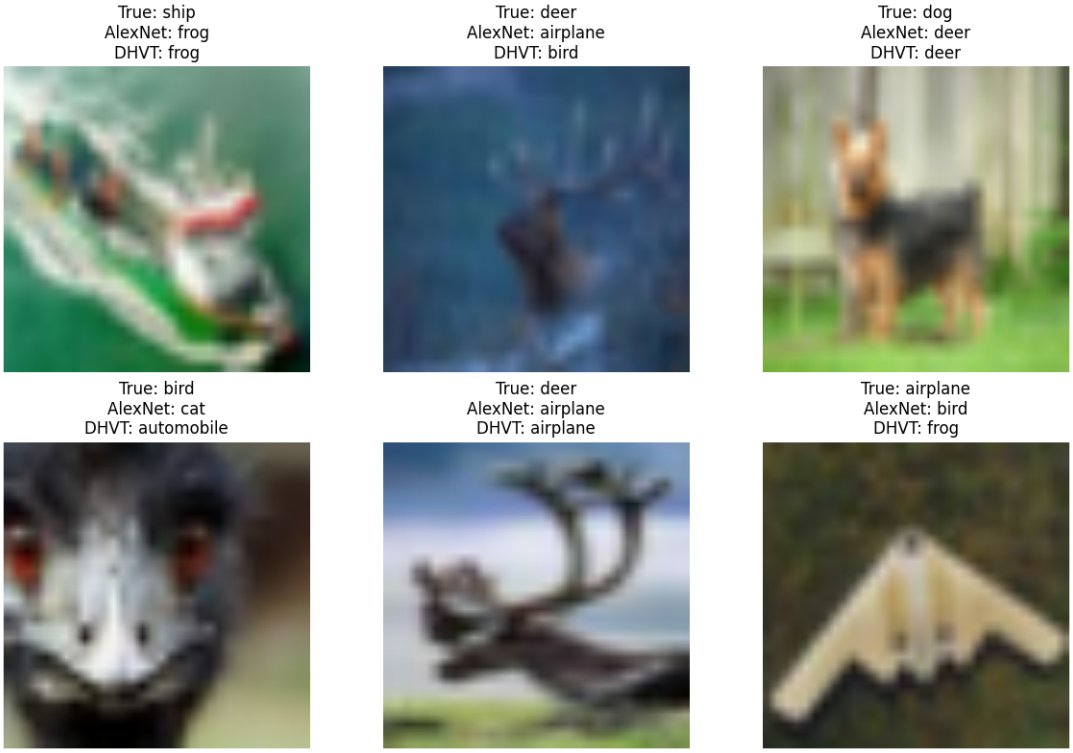

Figure 5: 04. Examples misclassified by both AlexNet and DHVT.

Figure 5 is especially informative because these are the hardest samples in the current evaluation. If both models fail on the same image, the reason is likely not architecture-specific only. These cases may correspond to ambiguous examples, low image quality, strong inter-class similarity, or samples that contain atypical object appearance.

From a report-writing perspective, this figure is important because it shifts the discussion from "which model is better" to "which inputs remain fundamentally difficult." This makes the analysis more balanced and more academically useful.

10.5 Confidence Distribution

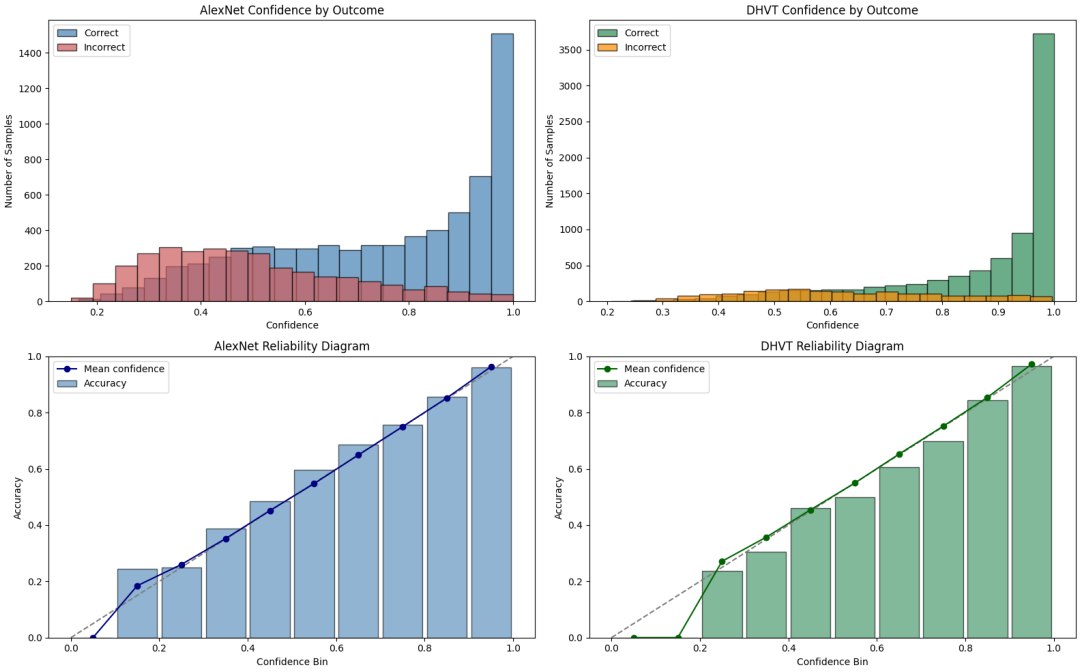

Figure 6: 05. Confidence distribution for AlexNet and DHVT, including correct and incorrect predictions.

Figure 6 is important because it moves beyond accuracy alone and examines how certain each model is about its predictions. DHVT shows a higher mean confidence, but the figure is useful mainly because it separates confidence on correct predictions from confidence on incorrect predictions. This distinction allows the report to discuss reliability rather than only raw performance.

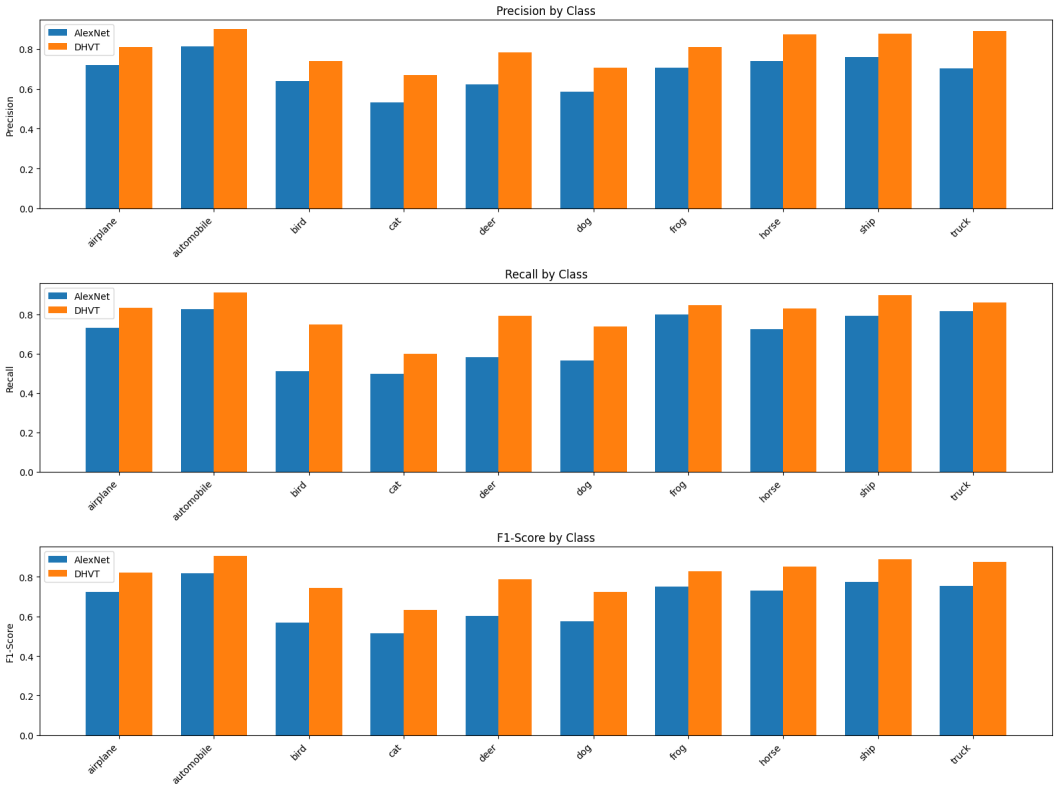

Figure 7: 08. Precision, recall, and F1-score comparison between AlexNet and DHVT.

Figure 7 complements the confusion matrix with class-aware summary metrics. The macro-level results are consistent with the rest of the analysis: DHVT is more precise, detects a larger proportion of the true samples, and maintains a better balance between precision and recall. This is important because it confirms that the advantage of DHVT is not caused by one-sided behavior such as being only conservative or only aggressive in its predictions.

10.6 Agreement and Disagreement Between Models

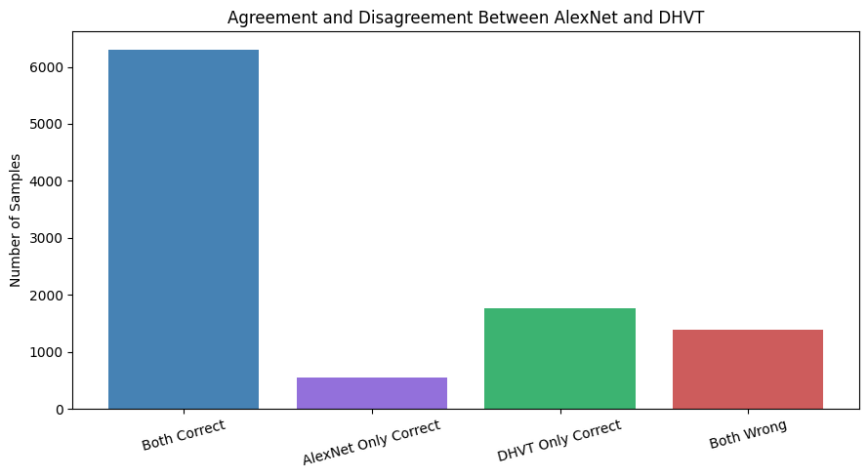

Figure 8: 06. Agreement and disagreement analysis between AlexNet and DHVT.

Figure 8 summarizes how often the two models succeed or fail on the same samples. The current notebook outputs are listed in Table 7.

Case	Ratio
Both models correct	0.6304
AlexNet only correct	0.0544
DHVT only correct	0.1758
Both models wrong	0.1394

Table 7: Agreement and disagreement ratios between AlexNet and DHVT.

This is one of the clearest pieces of evidence in the report. DHVT correctly classifies many samples that AlexNet misses, while the reverse case is relatively rare. This strongly supports the conclusion that DHVT is the stronger model in the present setup. At the same time, the non-zero "both wrong" portion confirms that there are still difficult samples for both architectures.

10.7 DHVT Attention Visualization

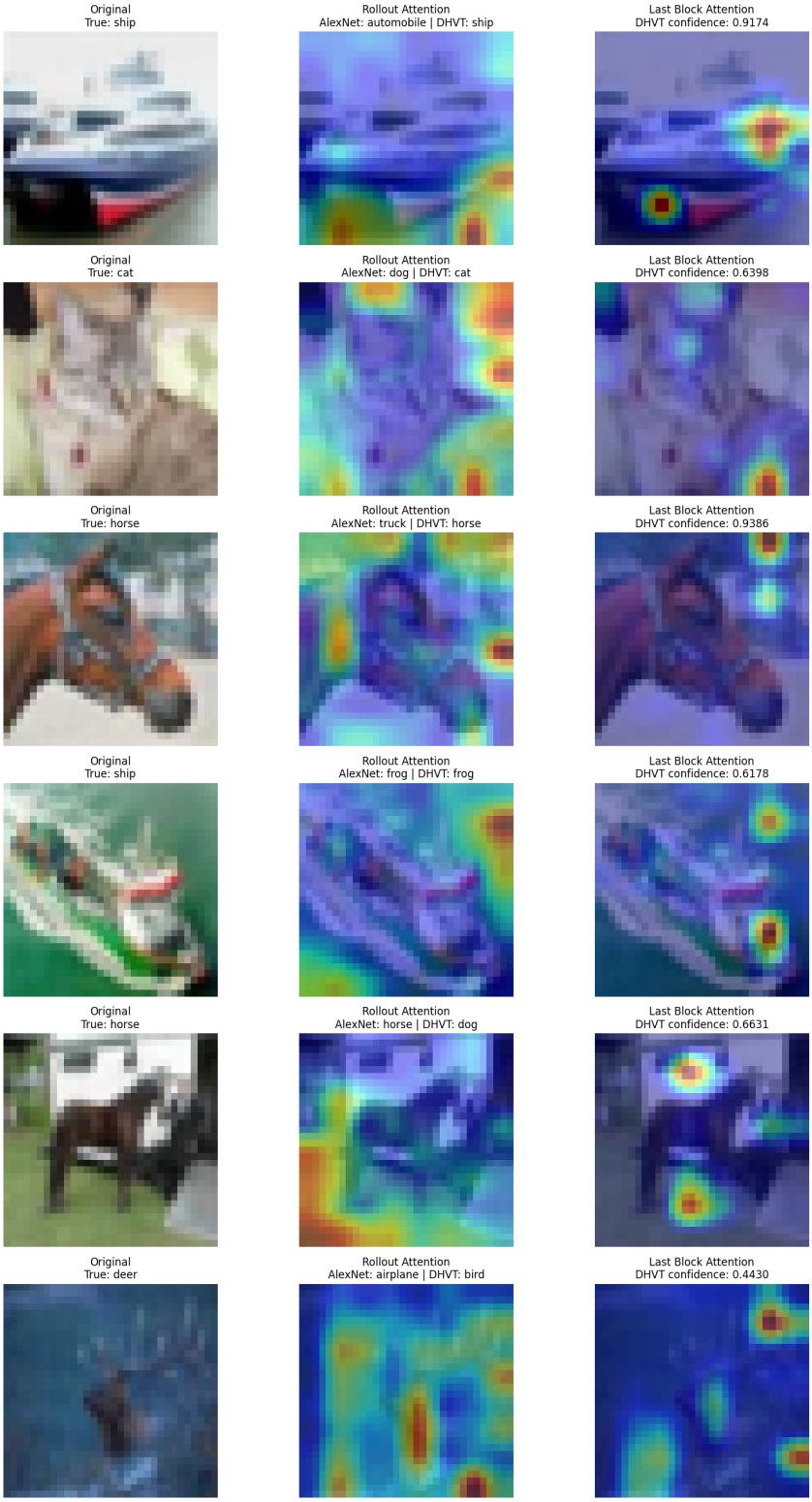

Figure 9: 09. DHVT attention visualization on representative validation samples.

Figure 9 adds an interpretability perspective to the comparison. While attention maps should not be treated as a complete explanation of model behavior, they still provide useful evidence about where the DHVT model focuses when it predicts a class. In several representative samples,

the attention is concentrated on the main object rather than being spread uniformly across the background, which is consistent with the stronger class-level results obtained by DHVT.

This figure is also pedagogically useful. The project is not only about determining which model wins on a benchmark, but also about showing how a transformer-style architecture differs from a classical CNN in the way it processes visual information. For that purpose, the attention visualization is one of the most informative additions to the report.

11 Discussion

Taken together, the visual and numerical evidence suggests that DHVT is the stronger model in this experiment. The advantage is visible at several levels. At the training level, the validation curves show better convergence and a higher final accuracy. At the global evaluation level, the macro precision, recall, and F1-score all favor DHVT. At the class level, the per-class accuracy plot shows that the improvement is broad rather than concentrated in a small number of categories.

This multi-level consistency is important. If the comparison relied only on one final accuracy value, the result would be less convincing. Here, however, the same conclusion appears in the learning curves, in the confusion matrices, in the per-class scores, and in the agreement analysis. In other words, DHVT does not win by chance or by exploiting one narrow metric; it performs better under several complementary views of evaluation.

The confidence analysis further refines the interpretation. DHVT is not only more accurate, but also slightly better calibrated according to the expected calibration error. This means that the transformer-style model is not simply more confident; its confidence is, on average, more aligned with its actual correctness. At the same time, the qualitative error examples and the shared failure cases show that both models still struggle on ambiguous or visually challenging samples. This prevents the discussion from becoming overly simplistic and keeps the analysis grounded in the actual limitations of the dataset.

However, the stronger result of DHVT in this experiment does not remove the weaknesses of transformer-style models. In general, transformer-based vision models are more demanding in design and tuning than CNNs. They often depend more strongly on data quantity, augmentation strategy, optimization details, and regularization choices. They are also usually less straightforward to explain from a classical feature-extraction perspective, and their attention visualizations, while useful, do not provide a complete causal explanation of the prediction process. Even when the parameter count is smaller than AlexNet in the present implementation, the transformer branch still remains architecturally more complex and less lightweight from an educational and implementation point of view.

AlexNet also has clear limitations. Its convolutional inductive bias is useful, but its representation is more local, and this can reduce its ability to model long-range visual relationships. In addition, the classical AlexNet design contains a very large classifier, which makes the model parameter-heavy in this project despite its older architecture. The results therefore suggest a balanced interpretation: AlexNet remains a strong, readable baseline, while DHVT offers better accuracy and class-level discrimination at the price of greater conceptual and implementation complexity.

Finally, the attention visualization provides a meaningful architectural explanation for the observed performance gap. The results suggest that DHVT benefits from its combination of local inductive bias and token-based global interaction. In the present experiment, this hybrid design appears to be a practical reason why the model achieves better discrimination on difficult

categories while remaining much smaller than the AlexNet baseline. Even so, this advantage should be understood as a result of the current setup rather than as proof that transformer-style models are always preferable in every image-classification problem.

12 Conclusion

This project compared AlexNet and DHVT on the same dataset under a shared and controlled experimental design. The notebook integrates training, checkpoint saving, checkpoint loading, custom-image inference, and several complementary visualization tools. This unified workflow makes the experiment appropriate both as a course project and as a reusable starting point for later comparisons on other datasets.

The final conclusion is clear: DHVT is the stronger model in the present setup. It achieves better validation accuracy, better macro precision, recall, and F1-score, and slightly better calibration, while using far fewer parameters than AlexNet. This makes the result particularly meaningful, because the improvement is not explained by a larger model but by a more effective architecture.

At the same time, the comparison should be interpreted with nuance. The present DHVT implementation performs better, but transformer-style models remain more complex, more sensitive to training design, and often less convenient to deploy or explain than classical CNNs. Conversely, AlexNet is easier to understand and still offers a strong baseline, but it is less competitive here and becomes surprisingly parameter-heavy because of its classifier structure.

The broader lesson of the project is therefore methodological as well as architectural. A fair model comparison should not stop at one aggregate number or at a list of advantages. Training curves, class-level metrics, confidence behavior, qualitative failure cases, and attention visualization all reveal different aspects of model behavior, including both strengths and weaknesses. By combining these views, the report shows not only that DHVT performs better in this experiment, but also what trade-offs come with that improvement.

A Appendix: Exported Figures Used in This Report

The figures inserted into this report were extracted directly from the notebook outputs and copied into the `latex/figures` directory with continuous numbering, as summarized in Table 8.

No.	File name
01	`01_validation_curves.png`
02	`02_confusion_matrices.png`
03	`03_misclassified_examples.png`
04	`04_both_models_wrong_examples.png`
05	`05_confidence_distribution.png`
06	`06_agreement_analysis.png`
07	`07_per_class_accuracy.png`
08	`08_precision_recall_f1.png`
09	`09_dhvt_attention_visualization.png`

Table 8: Exported figure files used in the report.

B References

1. Krizhevsky, A., Sutskever, I., and Hinton, G. E. *ImageNet Classification with Deep Convolutional Neural Networks*. 2012.

2. Dosovitskiy, A. et al. *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. 2021.

3. Lu, Y. et al. *Bridging the Gap Between Vision Transformers and Convolutional Neural Networks on Small Datasets*. 2022.

4. PyTorch Documentation. https://pytorch.org/

5. Torchvision Documentation. https://pytorch.org/vision/stable/